

Hanisha Mendu

Data Scientist — Brighton, UK

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Open to Relocation-Full Time Uk Work Permit

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PROFESSIONAL SUMMARY

Data Scientist with an MSc in Data Science and strong experience in machine learning, statistical modelling, and data-driven problem solving. Proficient in Python and SQL with hands-on experience building end-to-end ML pipelines, feature engineering workflows, and predictive models on real-world datasets. Experienced in translating complex data into actionable insights through modelling, experimentation, and visualisation. Passionate about applying machine learning techniques to solve business problems and uncover meaningful patterns in large-scale data.

CORE SKILLS

- Languages: Python, SQL, R (familiar)
- Machine Learning: Regression, Classification, Clustering, Model Evaluation, Feature Engineering
- Deep Learning: CNNs, Autoencoders, PyTorch, TensorFlow (familiar)
- Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Statistical Analysis
- Data Visualisation: Matplotlib, Seaborn, Power BI, t-SNE, UMAP
- Databases: MySQL, query optimisation, data extraction
- MLOps & Workflow: ML pipelines, model validation, experimentation workflows
- Concepts: Probability, Statistics, Hypothesis Testing, Model Generalisation

WORK EXPERIENCE

Wipro Limited

Data Analyst

Hyderabad, India

Aug 2021 – Aug 2024

- Performed exploratory data analysis (EDA) on large-scale business datasets to identify trends, anomalies, and actionable insights.
- Developed and optimised SQL queries to extract and transform data for analytical and modelling purposes.
- Built structured datasets and engineered features to support data-driven decision-making and predictive analysis.
- Designed automated data workflows improving data accessibility and reducing manual reporting effort.
- Implemented data validation checks ensuring high data quality for downstream analytics and modelling.
- Collaborated with cross-functional stakeholders to translate business problems into data-driven analytical solutions.
- Delivered dashboards and reports in Power BI to communicate insights effectively to non-technical stakeholders.

PROJECTS

Self-Supervised Learning of Animal Movement — MSc Dissertation

- Developed a self-supervised learning framework using a 1D CNN Autoencoder to learn latent representations from 3D trajectory data.
- Compressed 360-dimensional input sequences into 128-dimensional embeddings achieving efficient representation learning (validation loss: 0.0217).
- Applied MiniBatchKMeans clustering on 540,000+ samples, identifying 23 distinct behavioural movement patterns.
- Engineered domain-relevant features including velocity, angular change, and trajectory dynamics to enhance model learning.
- Evaluated embedding quality using t-SNE, UMAP, and cosine similarity to validate cluster separability.
- Built a complete ML pipeline including preprocessing, feature engineering, model training, and evaluation.

Geospatial Crop Yield Prediction using Multi-Source Data

- Developed a machine learning pipeline integrating 17 environmental datasets to predict crop yield across regions.
- Trained an MLP regression model (256-128 architecture) achieving $R^2 \approx 0.95$, demonstrating strong predictive performance.
- Performed feature engineering and data aggregation to improve model generalisation across temporal and spatial dimensions.
- Conducted model evaluation and validation to ensure robustness and minimise overfitting.
- Translated model outputs into interpretable insights for decision-making.

EDUCATION

University of Sussex <i>MSc Data Science</i>	<i>Sept 2024 – Sept 2025</i> Grade: Merit
Vignan Institute of Engineering for Women <i>B.Tech in Electronics and Communication</i>	<i>Jun 2017 – Jul 2021</i> GPA: 7.8/10

CERTIFICATIONS

- Data Analytics — KPMG (Forage)
- Agile Virtual Experience — JP Morgan & Chase
- Project Management — Accenture & CBRE
- Snowflake Fundamentals — Wipro
- Java Programming